MIL-C-915/46A
1 August 1972
SUPERSEDING 1/
MIL-C-915/46 (SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 2SWAU

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (Watertight)

First - Copper conductor, ASTM B-286 size 22-7, uncoated. Maximum diameter 0.033 (A) inch.
Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Minimum (A) thickness 0.010 inch. Colored insulation, one black and one white conductor for each pair.
Third - Clear polyamide jacket on each conductor, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch.
Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form a pair.
Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of (A) 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness of 0.003 inch. Standard Identification Code, applied by method 2.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, (A) sealed. Standard Identification Code, applied by method 2, on inner tape.
Eighth - The required number of pairs (see details) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with No. 1, from center outward. Non-fibrous fillers shall be employed to form a firm, well-rounded assembly.
Ninth - Binder tape applied helically with overlap.
Tenth - Standard thermoplastic jacket (gray). (See details for thickness.) Cable (A) surface marking required.

Details

(A)

Type and size	Number of pairs	Cable jacket thick. min. (inch)	Overall diameter	
			Min. (inches)	Max. (inches)
2SWAU-3	3	0.040	0.480	0.520
2SWAU-7	7	.040	.610	.660
2SWAU-10	10	.040	.770	.830
2SWAU-14	14	.050	.860	.930
2SWAU-19	19	.050	.970	1.040
2SWAU-24	24	.050	1.120	1.210
2SWAU-30	30	.050	1.190	1.280
2SWAU-37	37	.050	1.290	1.380
2SWAU-44	44	.050	1.460	1.550
2SWAU-61	61	.060	1.660	1.740

1/ Requirements formerly covered on MIL-C-24145/6.

(A) Denotes changes

FSC 6145

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - At 25°C., ohms/1000 ft., maximum	17.37
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Group A:	
Visual and dimensional	
Hydrostatic (open end); Leakage at 300 psi, 2 hrs., cu. inch, maximum	1.0
Capacitance	
Mutual capacitance - At 1 mHz, pF/ft., maximum	30
Capacitance unbalance - Percent, maximum	8
Characteristic impedance - At 1 mHz, ohms (one test for each 5000 ft. or fraction thereof)	75 ± 5
Group B:	
Attenuation - At 3 mHz, db/100 ft., maximum	3
Drip - At 75° ± 1°C.,	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Pulse Response time - Microseconds/500 ft., maximum	400
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum.	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum.	1400
Elongation - Percent, minimum.	50
Permanence of printing (jacket) - Cycles, minimum	250
Cable filler removability	

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cable aging and compatibility (95° ± 3°C.)	No failure

UNIT ORDERING LENGTHS:

Sizes	Feet (nominal)
3 through 30	1000
37 through 61	500

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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